Keeping Both Sides Equal

An equation is a balance. Whatever you do to one side, you have to do to the other — that single idea is the key to everything this week.

What Is an Equation?

An equation is a statement that two expressions are equal. Think of it as a balance scale: both sides have to stay level. Whatever operation you apply to one side, you must apply to the other, or the equation is no longer balanced.

Inverse Operations

To isolate a variable, undo whatever is currently being done to it — using the *inverse* of that operation, not just any operation.

Addition ↔ **Subtraction**
Multiplication ↔ **Division**

To solve $x + 7 = 12$, subtract 7 — not multiply by something. Use the inverse of what's actually there.

Solving One-Step Equations

$x + 9 = 15$ → subtract 9 → $x = 6$
$x - 4 = 10$ → add 4 → $x = 14$
$4x = 28$ → divide by 4 → $x = 7$
$x \div 3 = 6$ → multiply by 3 → $x = 18$

Solving Two-Step Equations

Undo operations in *reverse* order: addition/subtraction first, then multiplication/division — not the order the equation was built in.

$3x + 5 = 20$: subtract 5 → $3x = 15$. Divide by 3 → $\mathbf{x = 5}$.

A negative coefficient doesn't need a different method — $-2x + 9 = 1$ solves the same way, just using the Week 3 sign rules when you divide.

Combining Like Terms and Distributing First

Combine first: $2x + 3x + 4 = 19 \rightarrow 5x + 4 = 19 \rightarrow 5x = 15 \rightarrow x = 3$.

Distribute first: $3(x + 2) = 21 \rightarrow 3x + 6 = 21 \rightarrow 3x = 15 \rightarrow x = 5$.

Combining like terms simplifies an equation without changing its meaning or its solution. Distributing first is more reliable than trying to work around the parentheses, especially once there are more terms involved.

Checking Your Solution

Checking a solution isn't re-solving it — it's substituting your answer back into the *original* equation and confirming both sides come out equal.

Once x is found, a word problem isn't automatically finished — interpret the value in context, with the right units, before calling it the final answer. And a solution doesn't have to be a whole number: a fraction or decimal answer isn't automatically a mistake.

Equations in Real Life

Build the equation from the scenario, solve it, then state what the answer means.

A rental costs $40 plus $15 per day. Total cost was $115. How many days? 15d + 40 = 115 → 15d = 75 → **d = 5 days**.

Does My Answer Make Sense?

Check	Ask
Size	Is the result approximately what I expected?
Sign	Should it be positive or negative?
Substitution Check	Does plugging it back into the original equation balance?
Context	Does this value actually make sense in the situation?

Error Detective

Knowledge — “I don't remember which operation undoes this.”

Fix: review inverse operations — addition/subtraction, multiplication/division.

Process — undoing operations in the wrong order

In 3x + 5 = 20, dividing by 3 before subtracting 5. Fix: undo addition/subtraction first, then multiplication/division.

Execution — correct steps, arithmetic slip

The method was right; a sign or digit got dropped along the way. Fix: check the work, verify by substituting back in.

Comprehension — only changing one side of the equation

Subtracting 4 from the left side but not the right. Fix: remember the balance — the same operation applies to both sides.

Strategy — skipping distribution on a messy equation

Trying to work around 3(x+2) = 21 without distributing first makes it easy to lose track of a term. Fix: distribute first when more than one term sits outside the parentheses.

Quick Reference

Equation: a statement that two expressions are equal.

Inverse operations: addition ↔ subtraction, multiplication ↔ division.

Two-step order: undo addition/subtraction first, then multiplication/division.

Checking: substitute your answer into the original equation.

Check your answer: Size → Sign → Substitution Check → Context.

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